

System Driven Hardware Design

- Windowing and the Discrete Fourier Transform (DFT) -

International Master of Science

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Until now:

- We reviewed the DFT. We sampled a sinewave and calculated the DFT.

Now:

- Assuming we sampled a sinewave paying attention at the Nyquist criteria. Are there other parameters we have to pay attention while we perform the DFT?
- Whats happening if we use sensor data for the DFT?

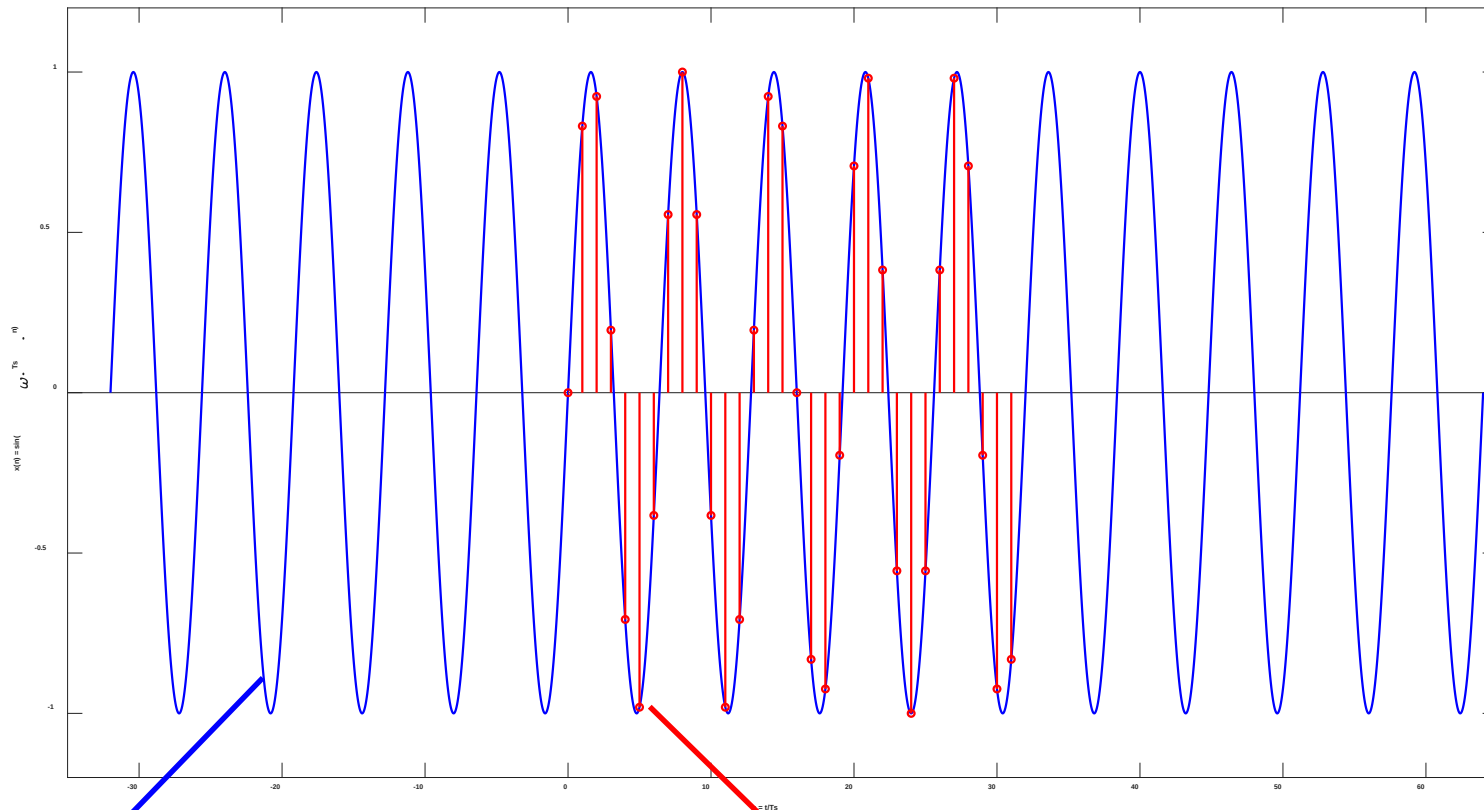


We look at windowing and its impact on spectra of the DFT.

Agenda

- **Sampling a periodic and time unlimited signal**
- Windowing and Leakage explained
- Comparison of different Windows
- Sum - Up

Until now: Taking some samples of a unlimited waveform



$\omega_0 = 2\pi f_p$	Angular frequency
f_p	Sine frequency
T_s	Sampling Time
$f_s = 1/T_s$	Sampling frequency
ϕ	Phase (assumed to be 0)
N	Number of samples

$$x(t) = A_0 \sin(\omega_0 t + \phi)$$

periodic and time unlimited

$$x(k) = A_0 \sin(\omega_0 T_s k + \phi)$$

non-periodic and time limited

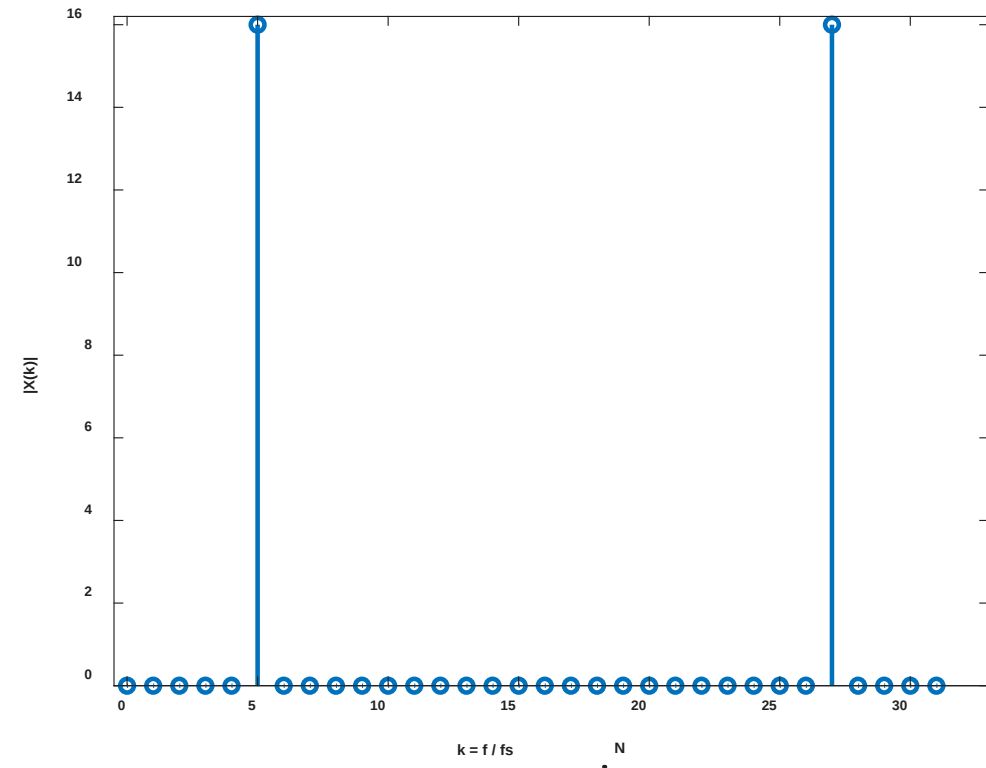
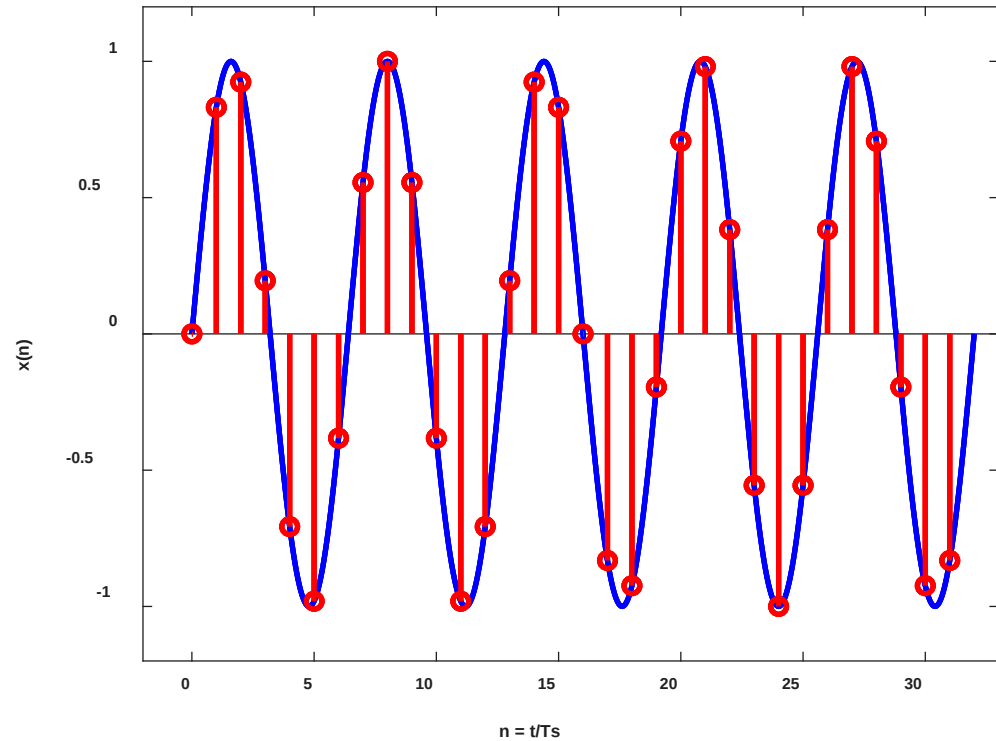
We obtain a series of N samples:

$$x(k) = A_0 \sin(2\pi \frac{f_p}{f_s} k) \text{ with } k \in [0, N - 1]$$

Performing DFT: Example A.)

A) $f_p = 0.15625$
 $f_s = 1$
 $N = 32$

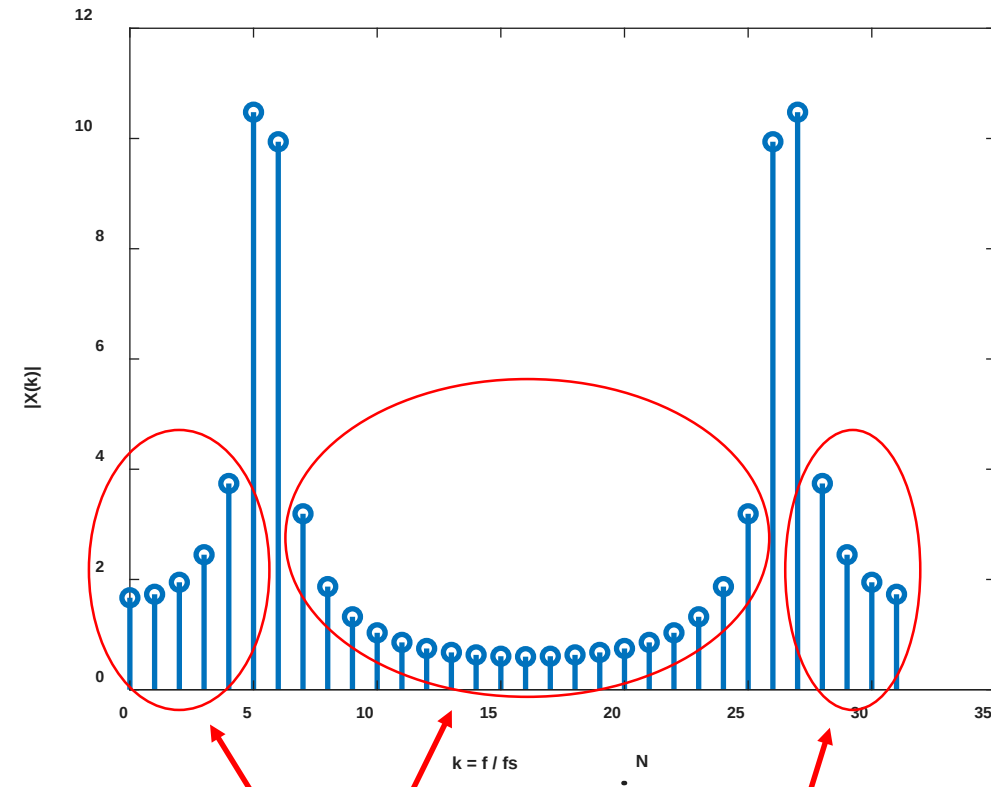
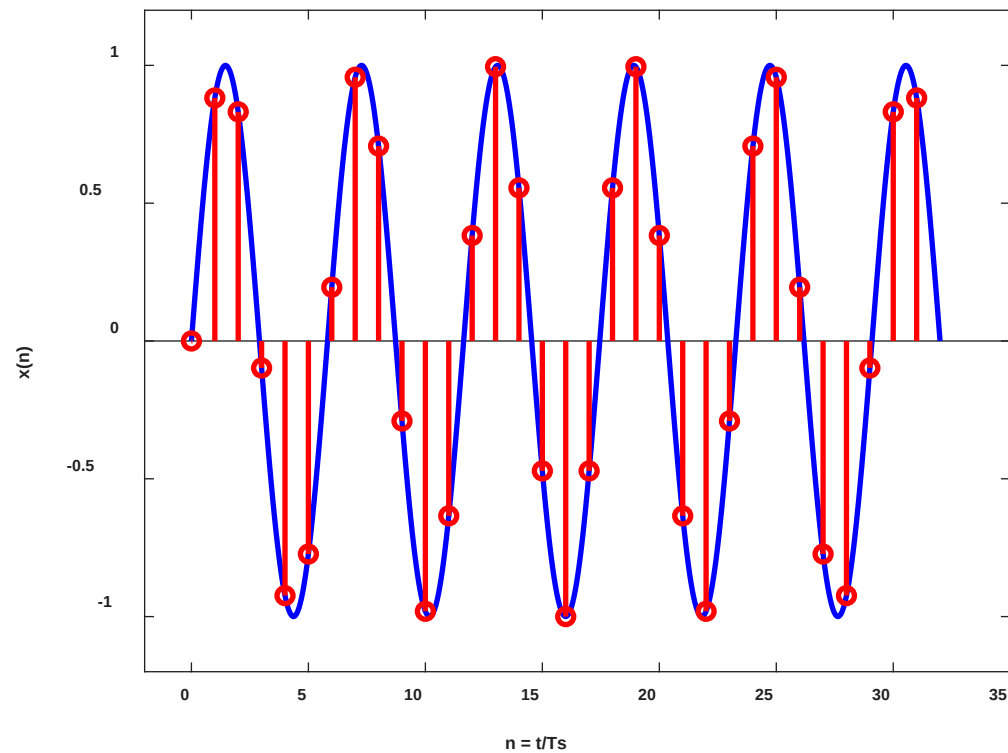
DFT:
$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi n}{N} k}$$



Performing DFT: Example B.)

B) fp = 0.171875
fs = 1
N = 32

$$\text{DFT: } X(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi n}{N} k}$$



Observation: Signal Leakage

Observation:

- 1.) We truncated (windowed) our periodic unlimited-time signal.
- 2.) The two signals have different frequencies

Example A.)

$$f_p / \left(\frac{f_s}{N} \right) = \text{[blue box]}$$

Bin hit.

Example B.)

$$f_p / \left(\frac{f_s}{N} \right) = \text{[blue box]}$$

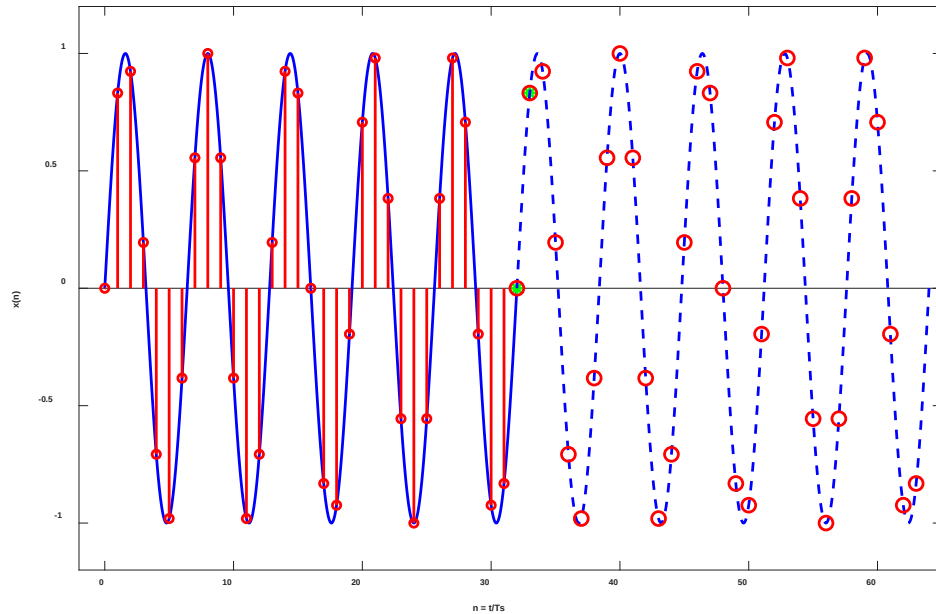
No Bin hit. Exactly between two bins.



What does a bin hit mean?

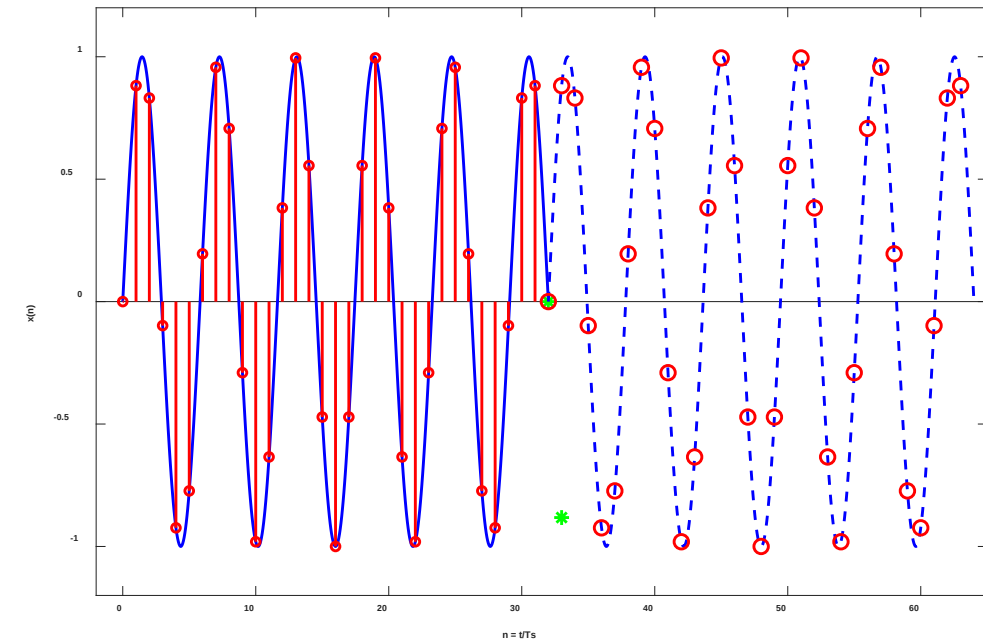
Example A.)

fp hits a bin,
means a periodic extension after the windowing.



Example B.)

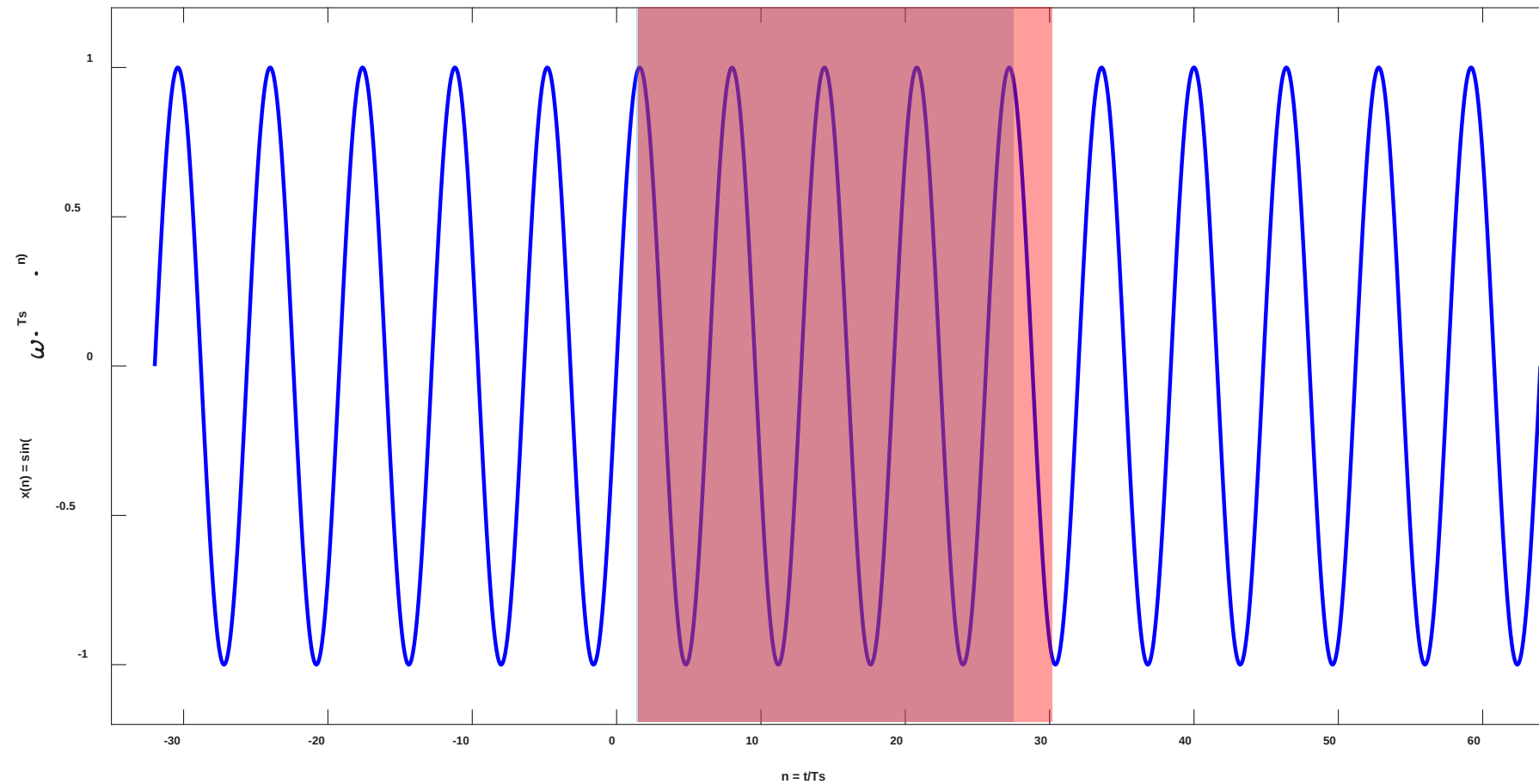
fp does not hit a bin,
means no periodic extension after the windowing.



If: $f_p T_s N = m \in \mathbb{N}$ holds, it means a **periodic extension** of the samples with **no leakage effect**.

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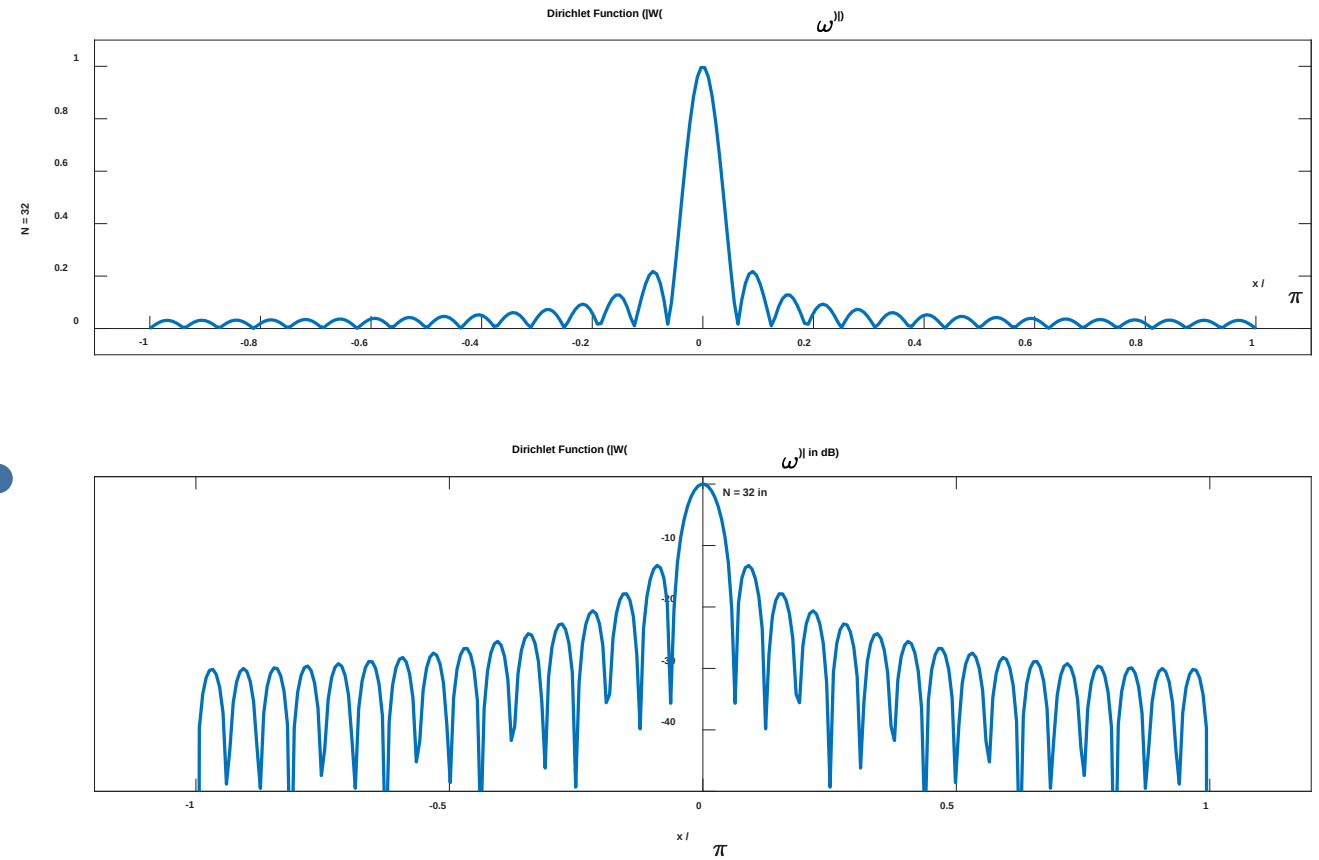
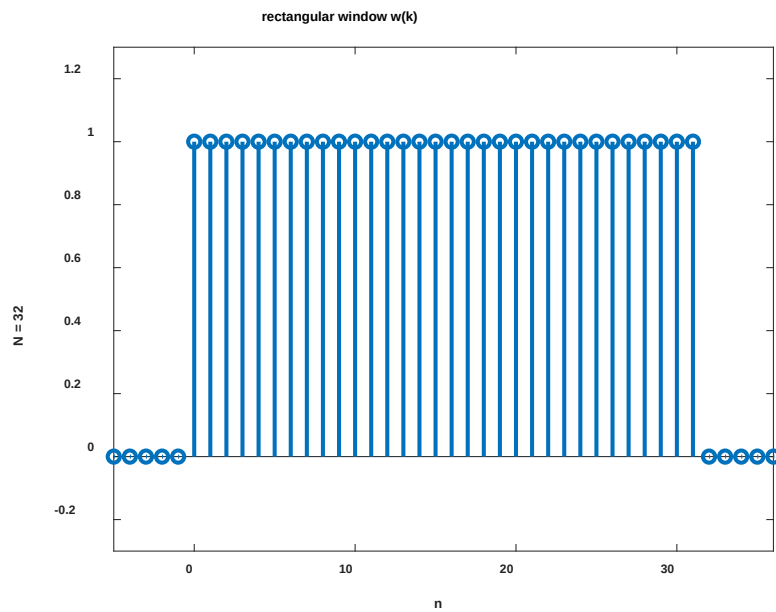


Blue: periodic extension **is** possible
Red: periodic extension **is not** possible

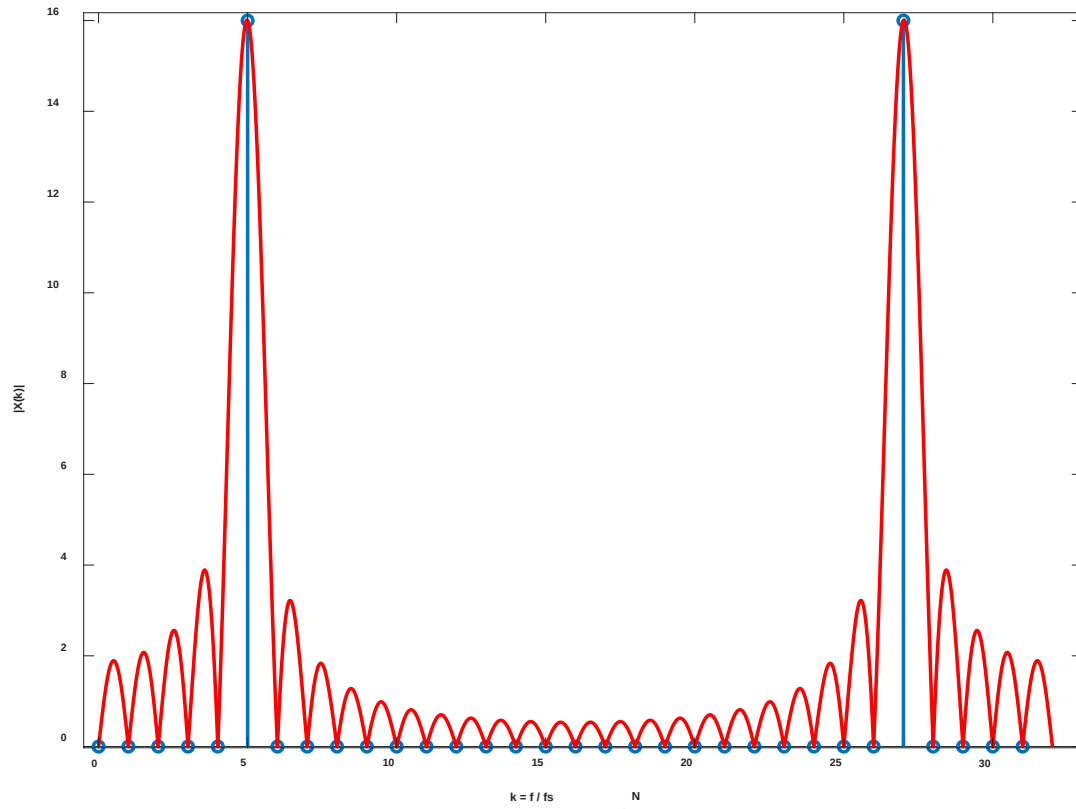


The question is not if, but how should we window the data?

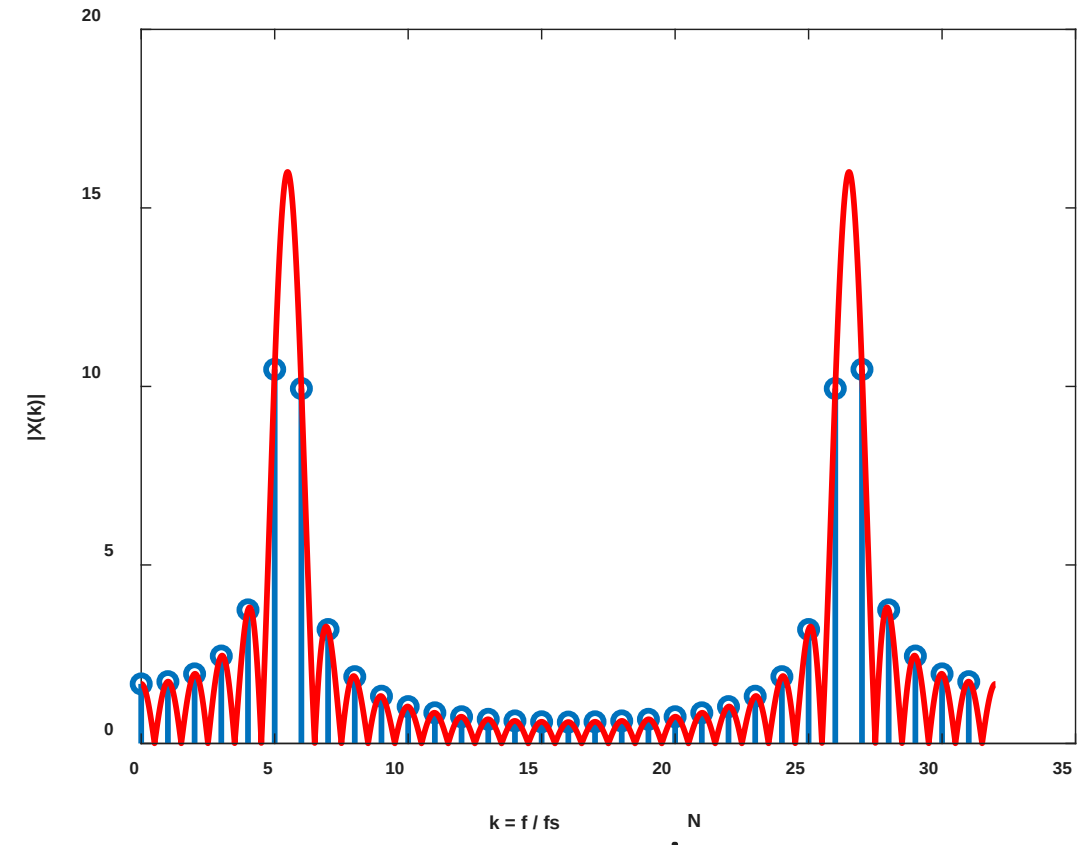
DTFT of the rectangular Window with N=32

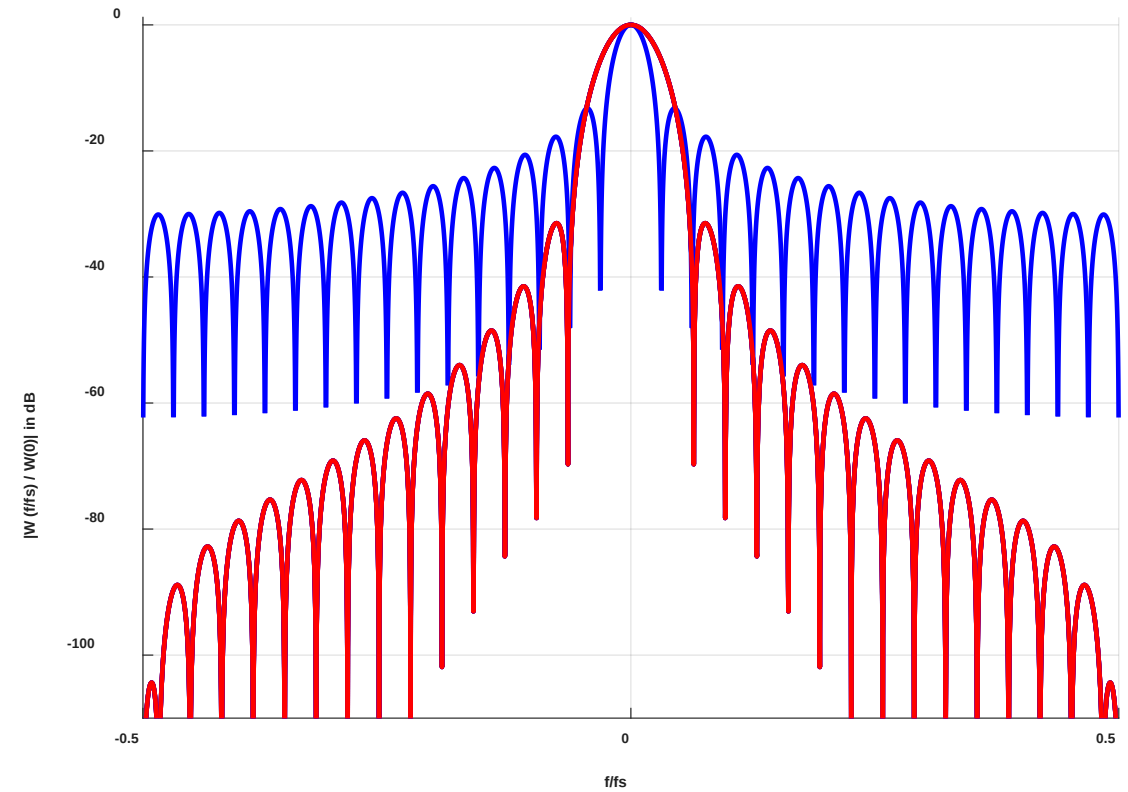
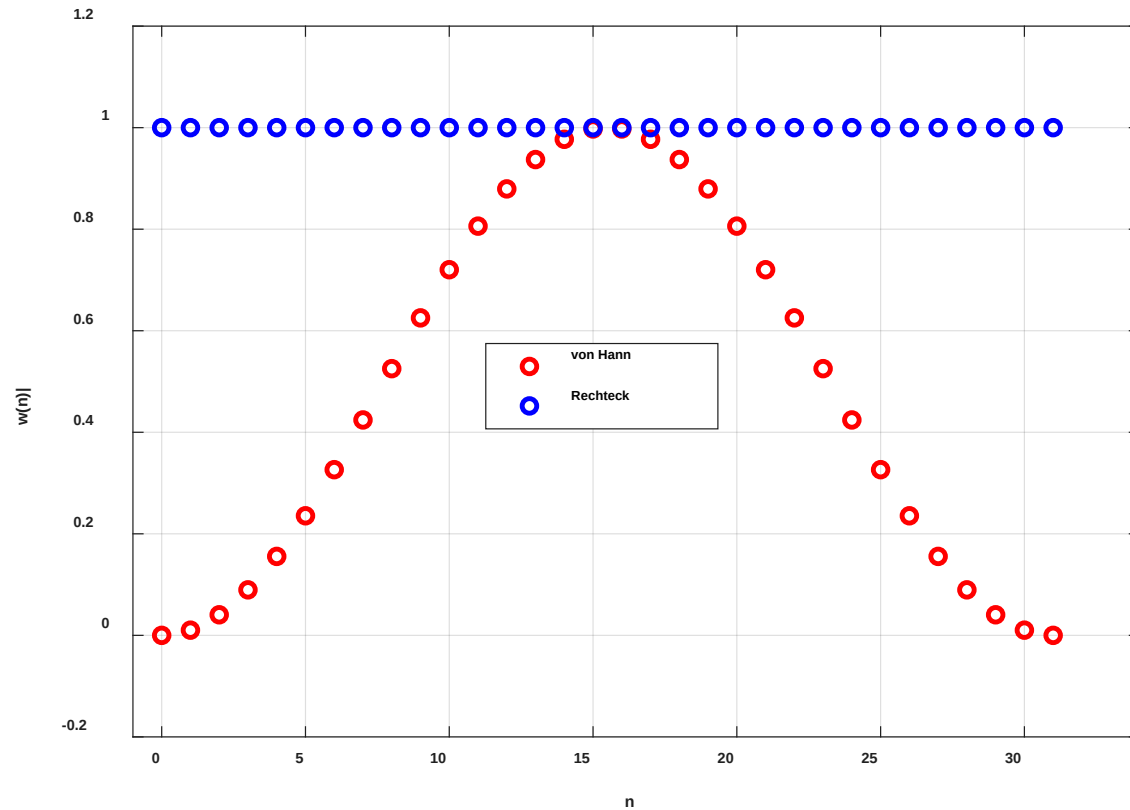


Example A.)

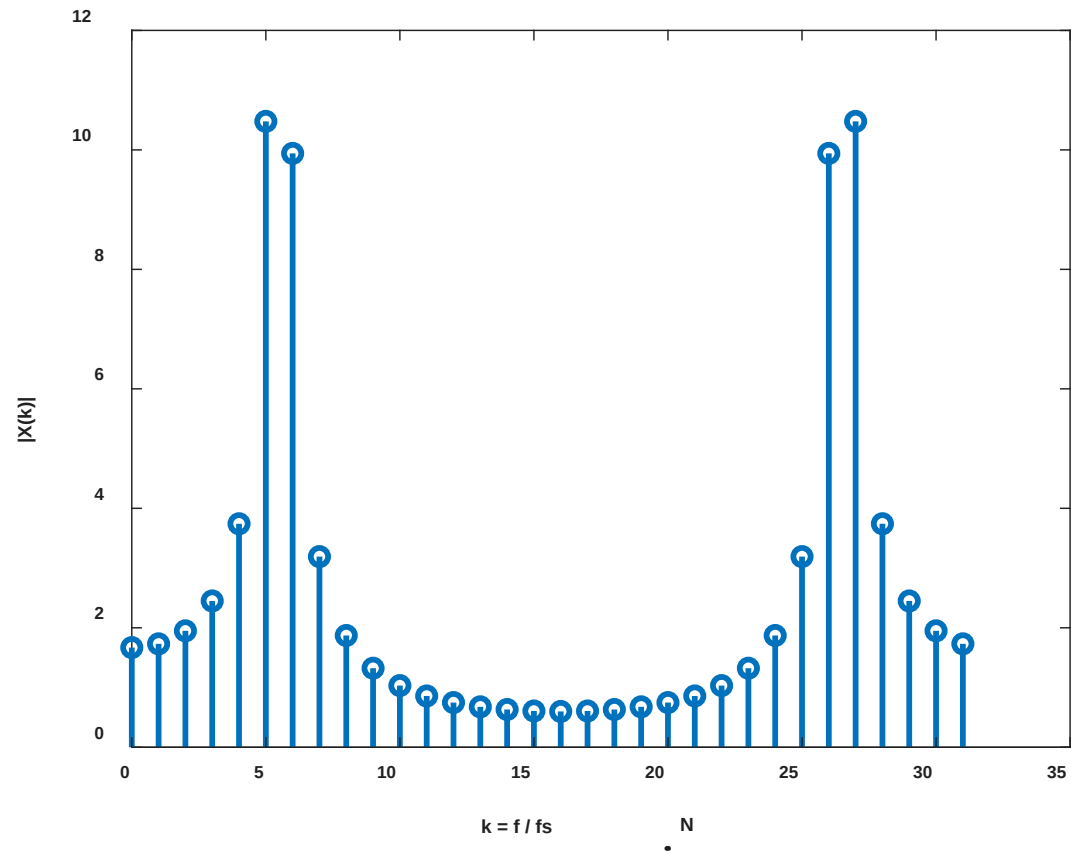


Example B.)

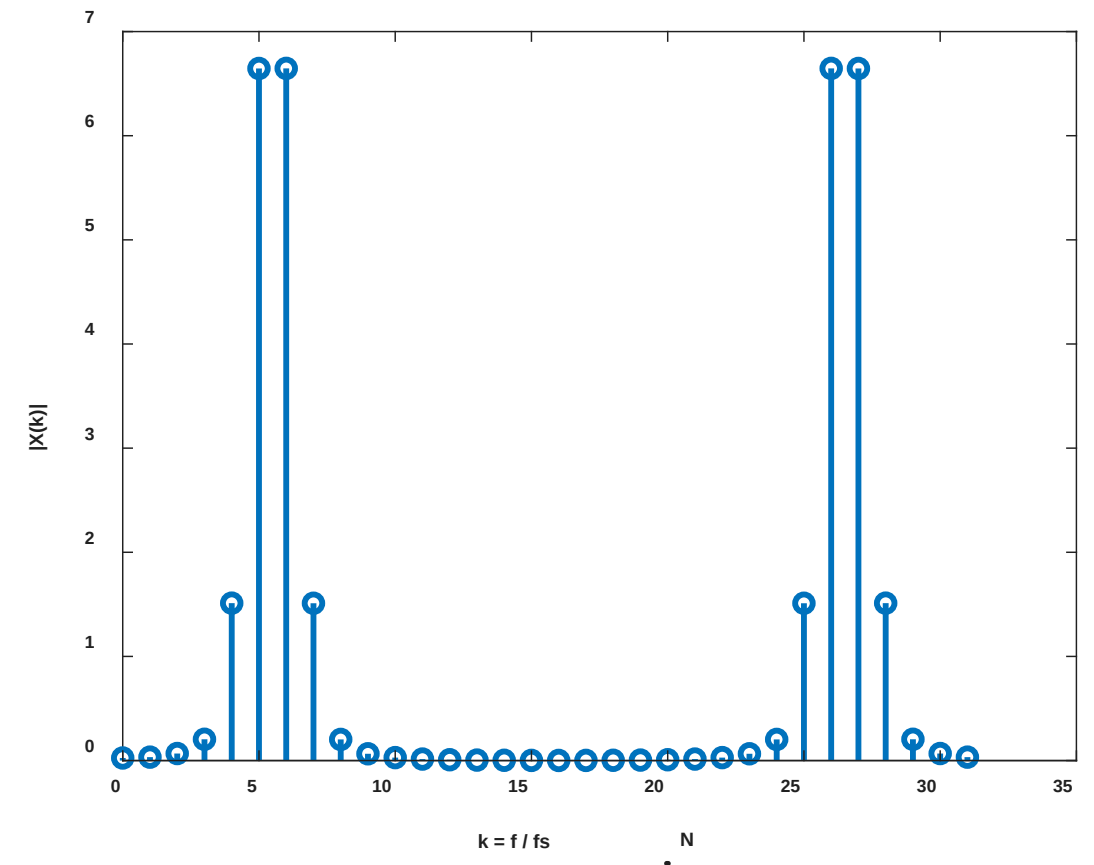




Rectangular Window



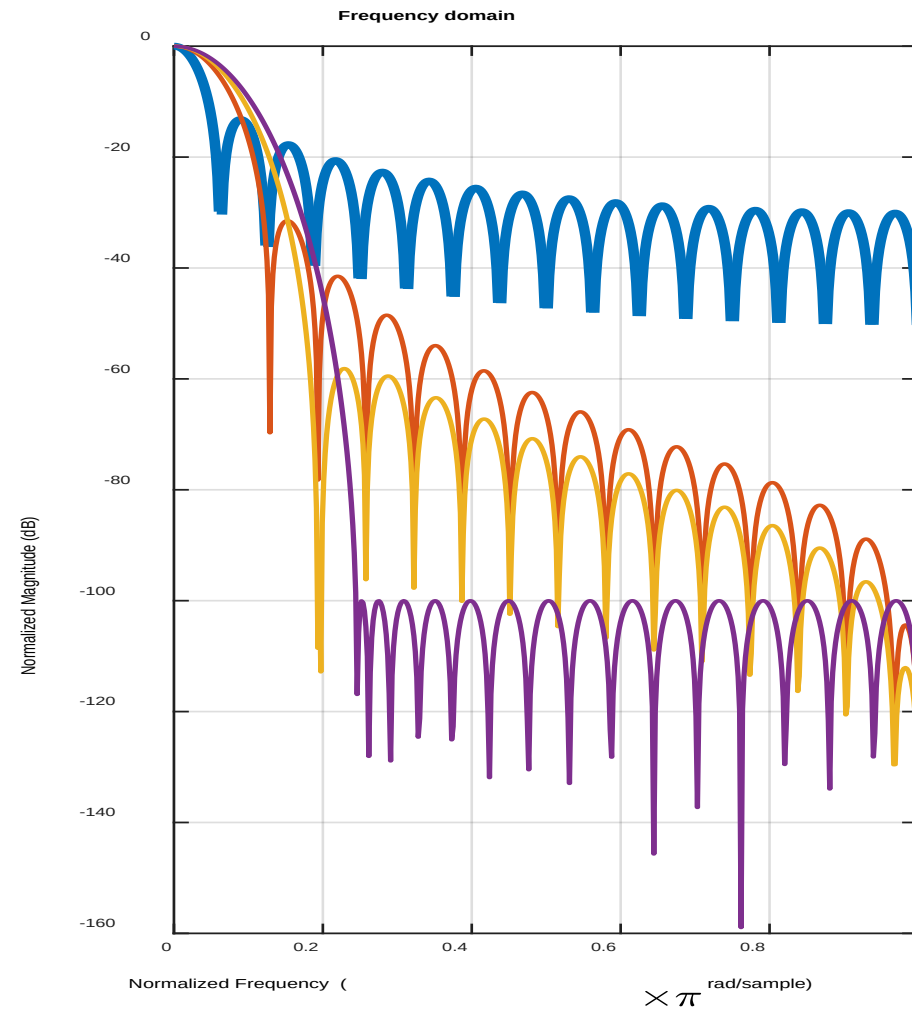
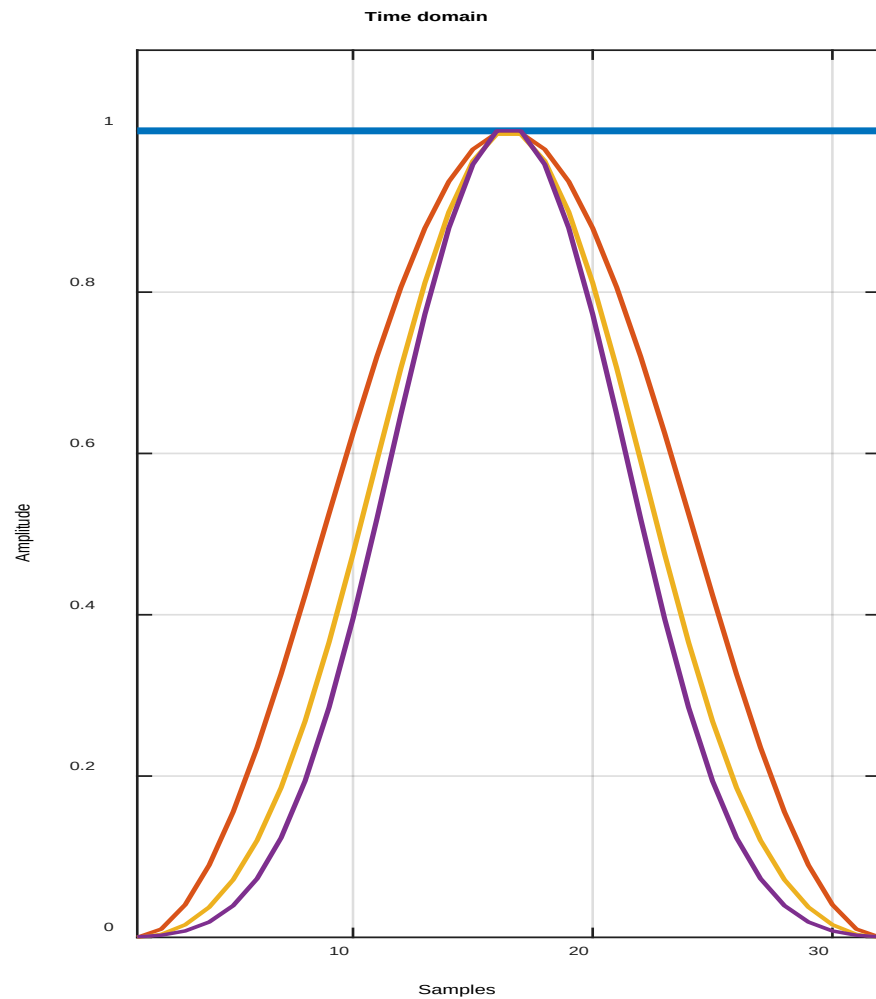
Van Hann Window



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Different Window Functions Compared (N=32)



Rectangular
Hanning ($\alpha=2$)
Blackman
Chebyshev

Matlab:
windowDesigner

Paper:
Fredric J. Harris
(1978)

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- All DFT assumes periodic extension of the sample series.

For sine waves this is obtain by:

$$f_p T_s N = m \in \mathbb{N}$$

- If periodic extension couldn't be guaranteed, a window different than rectangular should be used.
- Hence: the question is not if, but how should we window our time series.
- Windows main trade-off is: resolution vs. spectral leakage vs. sidelobes



Hints for
Your
Project

Run the Matlabsript “**Exercise_sine_windowing_student.m**” and do the exercises.

→ Once you understood the signal processing in Matlab, program it into the PSoC.

- “*Understanding Delta-Sigma Data Converters*”, Wiley / IEEE Press, Schreier, -> Appendix A, Spectral Estimation
- “*Digitale Signalverarbeitung*”, Vieweg+Teubner, Kammeyer und Kroschel, -> Ch. 8.2 Spektralanalyse periodischer Signale
- “*On the Use of Windows for Harmonic Analysis with Discrete Fourier Transform*”, IEEE, Fredric J. Harris (1978) -> summary and comparison of windows

Youtube Video:

- “Windowing and the DTFT”, by Barry Van Veen